

SAT Unit 3.5

Probability

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Overview

1 Probability

2 Practice

Probability

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Definition of Probability

Probability is a measure of how likely an event is to occur. It is defined as:

$$P(\text{Event}) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$$

Key Properties:

- $0 \leq P(A) \leq 1$ for any event A
- $P(\text{certain event}) = 1$
- $P(\text{impossible event}) = 0$
- $P(\text{not } A) = 1 - P(A)$
- For mutually exclusive events A and B :

$$P(A \text{ or } B) = P(A) + P(B)$$

Simple Probability Examples

Example 1: A fair die is rolled. What is the probability of getting an even number?

Even numbers: $\{2, 4, 6\} \Rightarrow$ Favorable outcomes: 3

$$\text{Total outcomes: } 6 \Rightarrow P(\text{even}) = \frac{3}{6} = \frac{1}{2}$$

Example 2: A card is drawn from a standard deck. What is the probability of drawing a heart?

$$\text{Hearts: } 13, \quad \text{Total cards: } 52 \Rightarrow P(\text{heart}) = \frac{13}{52} = \frac{1}{4}$$

Conditional Probability (Concept)

Conditional Probability is the probability of event A occurring **given** that event B has occurred.

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

Key Points:

- $P(A | B)$ narrows the sample space to only outcomes where B happens.
- Only defined if $P(B) \neq 0$.

Visual:

Think of "restricting" the universe to B , and checking how many of those are also A

Probability - Score Distribution Table

The same 20 contestants, on each of 3 days, answered 5 questions in order to win a prize. Each contestant received 1 point for each correct answer. The number of contestants receiving a given score on each day is shown in the table below.

Score	Day 1	Day 2	Day 3	Total
5 out of 5	2	2	3	7
4 out of 5	3	3	3	9
3 out of 5	4	5	3	13
2 out of 5	6	5	4	15
1 out of 5	2	4	3	9
0 out of 5	3	1	2	6
Total	20	20	20	60

No contestant received the same score on two different days. If a contestant is selected at random, what is the probability that the selected contestant received a score of 5 on Day 2 or Day 3, given that the contestant received a score of 5 on one of the three days?

Answer: Conditional Probability

Problem Context:

- 20 contestants each day, over 3 days (no repeat scores).
- Table shows number of contestants scoring 0–5 on each day.
- Total scores of 5 across all days: $2 + 2 + 3 = 7$

Given: A contestant received a score of 5 on **one of the days**.

Find: Probability that this contestant scored 5 on Day 2 or Day 3.

Step 1: Count favorable outcomes

Day 2: 2 Day 3: 3 Total favorable = $2 + 3 = 5$

Step 2: Count total outcomes

Total contestants with score of 5: 7

Answer:

$$P(\text{Day 2 or Day 3} \mid \text{Score 5}) = \frac{5}{7}$$

Correct Answer:

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$\frac{5}{7}$

Practice

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Question: The table summarizes the distribution of age and...

The table summarizes the distribution of age and favorite sport for 150 members of a country club.

	Squash	Tennis	Golf	Total
30–44 years	18	10	22	50
45–59 years	11	14	25	50
60+ years	21	26	3	50
Total	50	50	50	150

One of these members will be selected at random. What is the probability of selecting a member whose favorite sport is tennis, given that the member is at least 45 years of age? (Express your answer as a decimal or fraction, not as a percent.)

Answer Explanation: The table summarizes the distribution of age and...

Step 1. Identify the relevant group:

$$\text{At least 45 years of age} = (45-59) + (60+) = 50 + 50 = 100.$$

Step 2. Find the total number of tennis players in that group:

$$14(45-59) + 26(60+) = 40.$$

Step 3. Calculate the probability:

$$P(\text{Tennis} \mid \text{At least 45}) = \frac{40}{100} = 0.4.$$

Answer: .

Question: A new medical test is designed to give a positive...

A new medical test is designed to give a positive indicator to a patient who has a certain virus and a negative indicator to a patient who does not. The table summarizes the results for 1,000 patients who have taken this test.

	Negative Indicator	Positive Indicator	Total
Has virus	30	370	400
Does not have virus	550	50	600
Total	580	420	1,000

Based on the table, what is the probability that the test gives the wrong indicator?

- A. 5%
- B. 8%
- C. 10%
- D. 12%

Answer Explanation: A new medical test is designed to give a positive...

Step 1. Count the number of wrong indicators:

- False negatives (Has virus, but test negative): 30
- False positives (Does not have virus, but test positive): 50

$$30 + 50 = 80$$

Step 2. Divide by the total number of patients:

$$\frac{80}{1,000} = 0.08$$

Step 3. Answer:

$$0.08 = 8\%$$

Correct Answer: B

Question: The table shows the results of a survey in which...

The table shows the results of a survey in which students at three colleges were asked whether they live on campus or off campus.

	Off campus	On campus	Total
College A	335	689	1,024
College B	124	482	606
College C	596	274	870
Total	1,055	1,445	2,500

If one of the students surveyed is selected at random, which of the following probabilities has the greatest value?

- A. The probability that the student lives off campus, given that the student attends College C.
- B. The probability that the student lives on campus, given that the student attends College C.
- C. The probability that the student attends College C, given that the student lives off campus.

Answer Explanation: The table shows the results of a survey in which...

Step 1. Calculate each conditional probability:

- A: $\frac{596}{870} \approx 0.685$
- B: $\frac{274}{870} \approx 0.315$
- C: $\frac{596}{1055} \approx 0.565$
- D: $\frac{274}{1445} \approx 0.19$

Step 2. Compare:

$$0.685 > 0.565 > 0.315 > 0.19$$

Correct Answer: A

Thank You!

We appreciate your time and focus.

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